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ASSIGNMENT 3

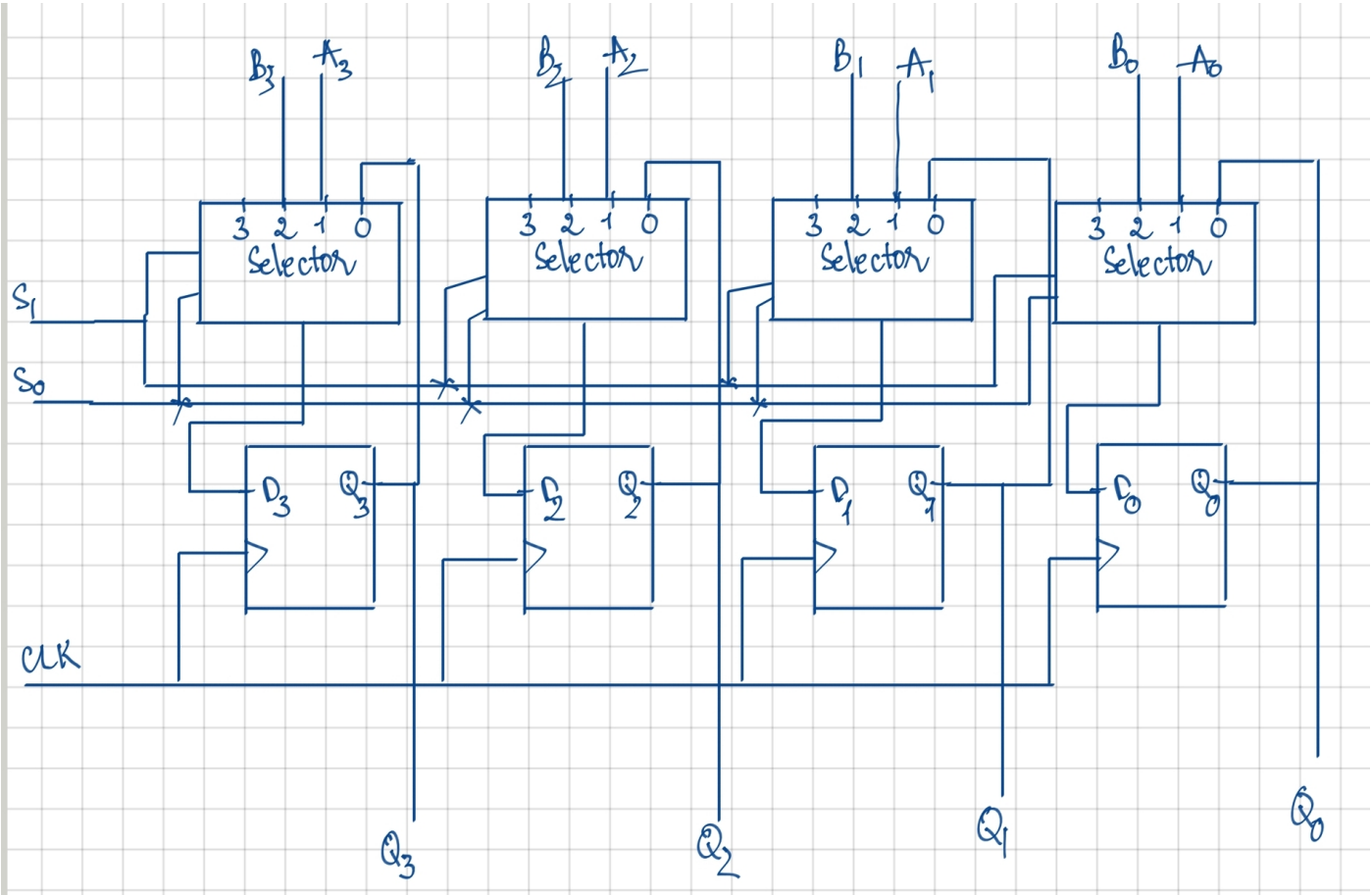
7.2, 7.3, 7.4, 7.5, 7.8 (dùng 4 bit), 7.12, 7.13 (English texthook)

7.2 **(Registers) Design a register with two load signals that enable the loading of data from two different sources.**

Solution:

Giả sử các nguồn đầu vào là dữ liệu 4 bit là A và B. Các chân tín hiệu chọn chế độ là S1, S0

S1	S0	Hành động
0	0	Không đổi
0	1	Load A
1	0	Load B
1	1	Không dùng



7.3 (Registers) Explain the difference between resetting the register and loading an all-0's input into the register.

Solutions:

- Reset thanh ghi là ép tất cả các output về 0 ngay lập tức bất kể output hiện tại bằng bao nhiêu thông qua việc tác động chân CLR là tín hiệu bất đồng bộ của các Flip Flop, bất kể xung Clock
- Nạp vào giá trị 0 vào thanh ghi là nạp vào thanh ghi dữ liệu 0 thông qua đường dữ liệu thông thường, output sẽ thay đổi theo xung Clock, khi nào enable Load được bật và có xung Clock hợp lệ thì mới tiến hành nạp.

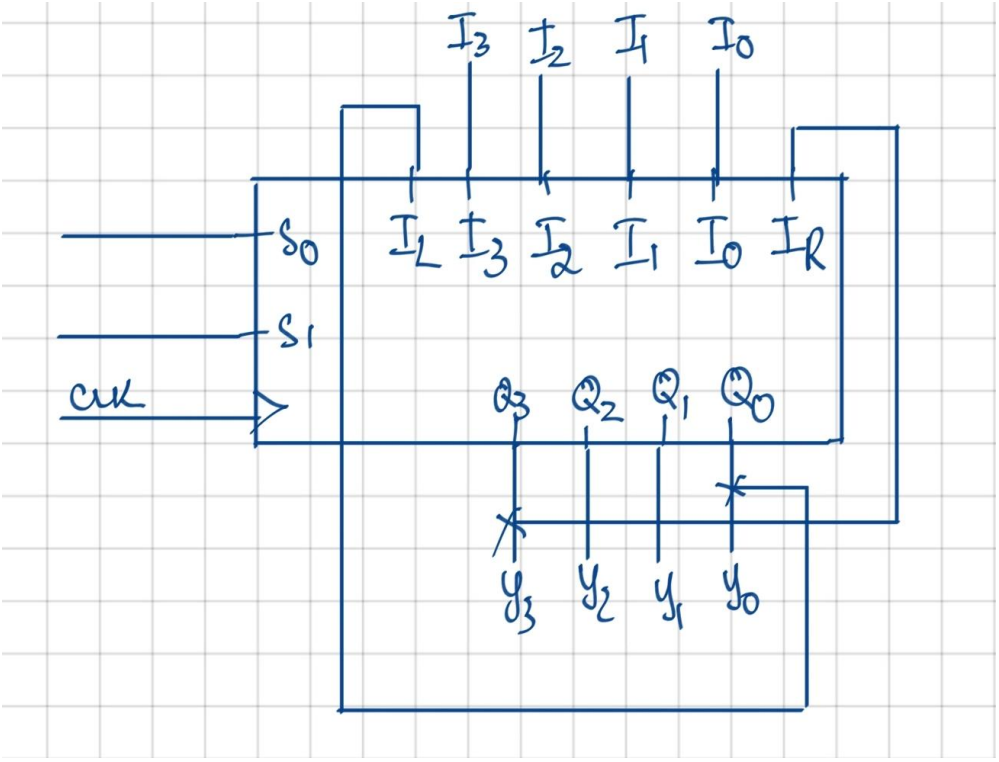
7.4 (Registers) Design a 16-bit register that can load new data and rotate or shift its content left or right.

S2	S1	S0	
0	0	0	Không đổi
0	0	1	Load
0	1	0	Dịch trái
0	1	1	Dịch phải
1	0	0	Xoay trái
1	0	1	Xoay phải

7.5 (Shift registers) Using a 4-bit shift register, construct a 4-bit register that can rotate its content one position to the left or right.

Solutions:

S1	S0		Output mong muốn xoay	Output của thanh ghi dịch 4 bit
0	0	Không đổi	Q3 Q2 Q1 Q0	Q3 Q2 Q1 Q0
0	1	Load	I3 I2 I1 I0	I3 I2 I1 I0
1	0	Xoay trái	Q2 Q1 Q0 Q3	Q2 Q1 Q0 QR
1	1	Xoay phải	Q0 Q3 Q2 Q1	QL Q3 Q2 Q1



7.8 (Counters) Design a binary counter that counts up only in:

(a) Even numbers (0, 2, 4, 6, 8, . . .)

(b) Odd numbers (1, 3, 5, 7, 9, . . .)

Solution:

a)

Q_3	Q_2	Q_1	Q_0	D_3	D_2	D_1	D_0	Q_3^+	Q_2^+	Q_1^+	Q_0^+
0	0	0	0	0	0	1	0	0	0	1	0
0	0	1	0	0	1	0	0	0	1	0	0
0	1	0	0	0	1	1	0	0	1	1	0
0	1	1	0	1	0	0	0	1	0	0	0
1	0	0	0	1	0	1	0	1	0	1	0
1	0	1	0	1	1	0	0	1	1	0	0
1	1	0	0	1	1	1	0	1	1	1	0
1	1	1	0	0	0	0	0	0	0	0	0

D_3	Q_1Q_0 00	01	11	10
Q_3Q_2 00	0	X	X	0
01	0	X	X	1
11	1	X	X	0
10	1	X	X	1

$$D_3 = Q_3Q_1' + Q_3Q_2' + Q_3'Q_2Q_1$$

D_1	Q_1Q_0 00	01	11	10
Q_3Q_2 00	1	X	X	0
01	1	X	X	0
11	1	X	X	0
10	1	X	X	0

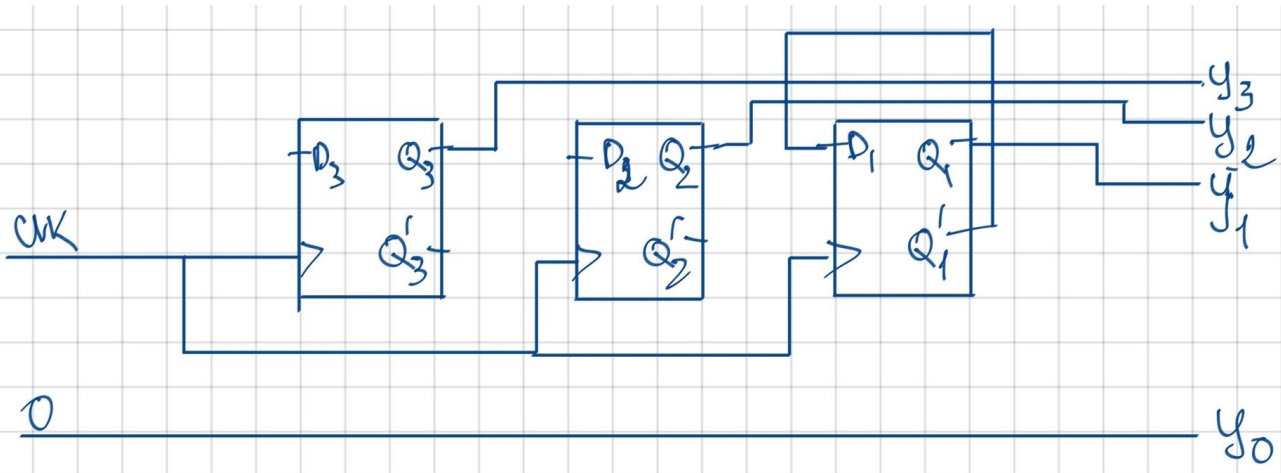
$$D_1 = Q_1'$$

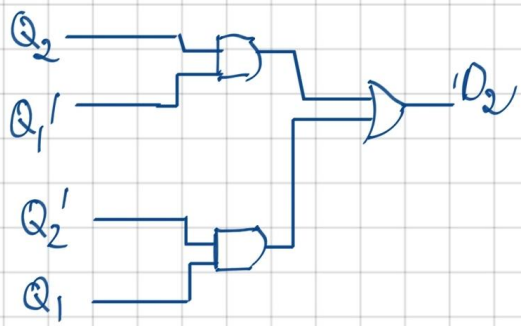
D_2	Q_1Q_0 00	01	11	10
Q_3Q_2 00	0	X	X	1
01	1	X	X	0
11	1	X	X	0
10	0	X	X	1

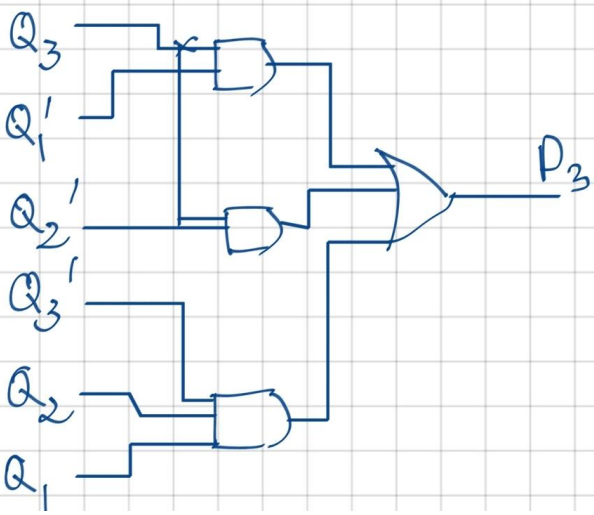
$$D_2 = Q_2Q_1' + Q_2'Q_1$$

D_0	Q_1Q_0 00	01	11	10
Q_3Q_2 00	0	X	X	0
01	0	X	X	0
11	0	X	X	0
10	0	X	X	0

$$D_0 = 0$$

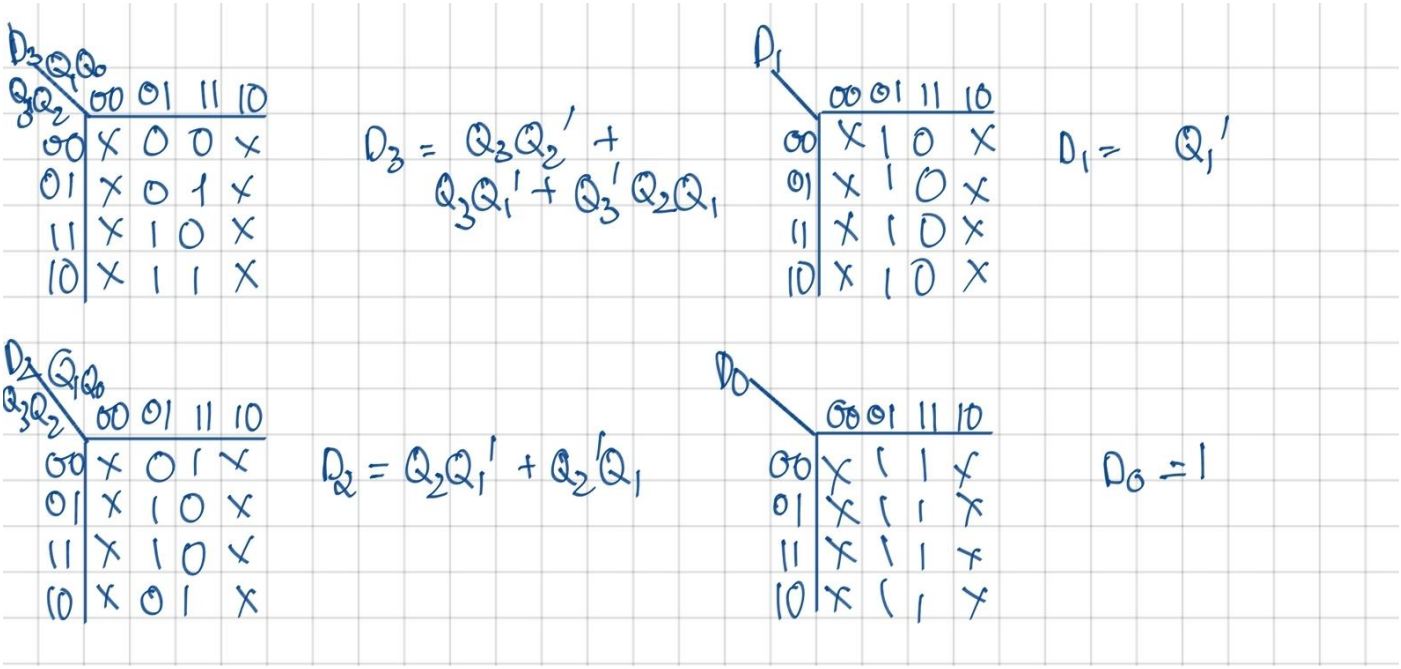


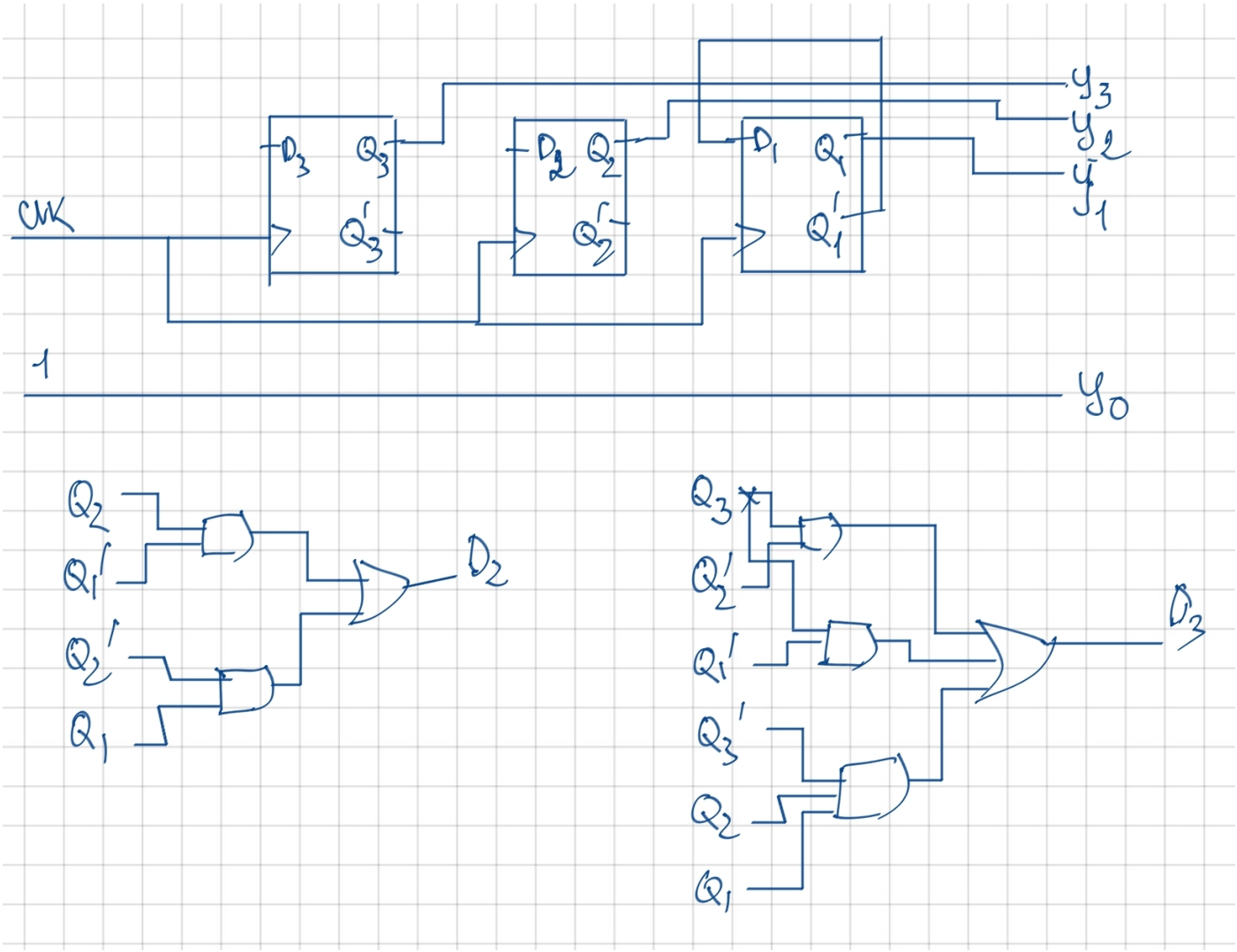




b)

Q_3	Q_2	Q_1	Q_0	D_3	D_2	D_1	D_0	Q_3^+	Q_2^+	Q_1^+	Q_0^+
0	0	0	1	0	0	1	1	0	0	1	1
0	0	1	1	0	1	0	1	0	1	0	1
0	1	0	1	0	1	1	1	0	1	1	1
0	1	1	1	1	0	0	1	1	0	0	1
1	0	0	1	1	0	1	1	1	0	1	1
1	0	1	1	1	1	0	1	1	1	0	1
1	1	0	1	1	1	1	1	1	1	1	1
1	1	1	1	0	0	0	1	0	0	0	1





7.12 (Counters) Design a counter that counts in the following sequence: 1, 5, 9, 11, 13, 15, 1, 5, ...

Q_3	Q_2	Q_1	Q_0	D_3	D_2	D_1	D_0	Q_3^+	Q_2^+	Q_1^+	Q_0^+
0	0	0	1	0	1	0	1	0	1	0	1
0	1	0	1	1	0	0	1	1	0	0	1
1	0	0	1	1	0	1	1	1	0	1	1
1	0	1	1	1	1	0	1	1	1	0	1
1	1	0	1	1	1	1	1	1	1	1	1
1	1	1	1	0	0	0	1	0	0	0	1

D_3

	00	01	11	10
00	X	0	X	X
01	X	1	X	X
11	X	1	0	X
10	X	1	1	X

$D_3 = Q_3 Q_2' + Q_2 Q_1'$

D_1

	00	01	11	10
00	X	0	X	X
01	X	0	X	X
11	X	1	0	X
10	X	1	0	X

$D_1 = Q_3 Q_1'$

D_2

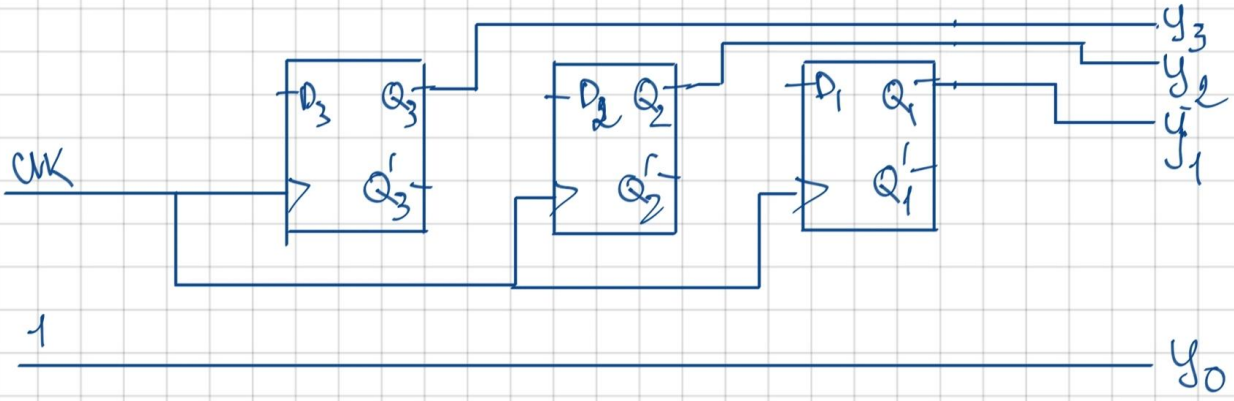
	00	01	11	10
00	X	1	X	X
01	X	0	X	X
11	X	1	0	X
10	X	0	1	X

$D_2 = Q_3 Q_2' + Q_2' Q_1 + Q_3 Q_2 Q_1'$

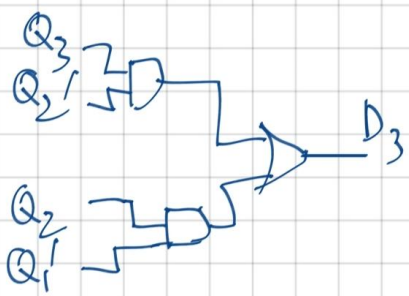
D_0

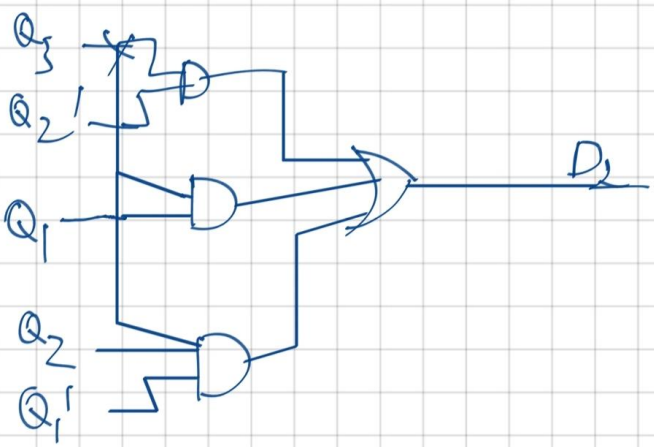
	00	01	11	10
00	X	1	X	X
01	X	1	X	X
11	X	1	1	X
10	X	1	1	X

$D_0 = 1$







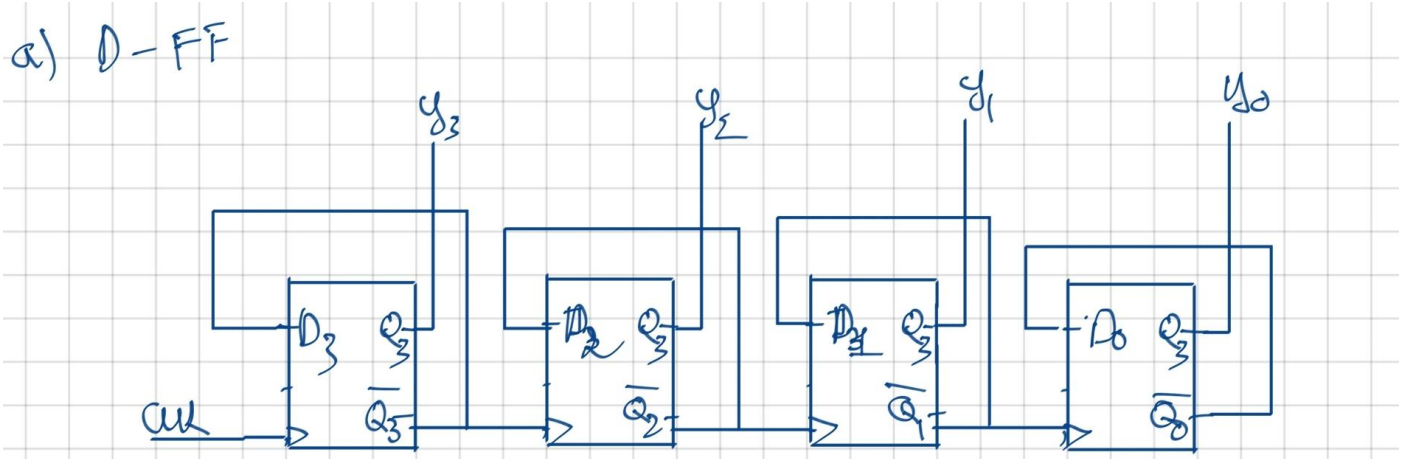


7.13 (Asynchronous counters) Construct a 4-bit asynchronous counter using:

(a) D flip-flops

(b) J-K flip-flops

a) D-FF



b) J-K FF

